

Spotify can predict your next emotion.

Reimagining Spotify with Generative AI to deliver personalised, emotionally intelligent music experiences that reduce churn and grow premium revenue.

602M

Monthly Active Users

31%

Churn within 90 days

68%

Skip AI-generated playlists



The Discovery Problem

Spotify's recommendation engine is powerful — but passive. It reacts to listening history, not to emotional state, context, or intent. Users in grief get upbeat playlists. Users in flow get mood-breaking transitions. The result: friction, skipping, and churn.

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Discovery Fatigue

Users spend avg. 8 min searching for the right playlist, often giving up and switching apps.

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Context Blindness

Recommendations ignore real-world context — weather, time, mood, physical activity.

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Churn at Free Tier

31% of free users churn within 90 days. AI-curated paywall features aren't compelling enough to convert.

User Segments

40% Passive Listeners

Play ambient/background music. Low engagement, high churn risk.

35% Active Explorers

Seek new artists & genres. Frustrated by repetition.

25% Mood Driven

Curate by emotion. Feel unheard by current AI playlists.

Traditional Approaches Fall Short

- ❌ Collaborative filtering: only mirrors past behaviour, no emotional intelligence
- ❌ Rule-based playlists: static, can't adapt to real-time context
- ❌ Predictive ML: forecasts what you listened to, can't understand what you need
- ❌ Manual curation: doesn't scale to 602M users across 100+ moods

GenAI Uniquely Enables

[v] Emotional Context Understanding

LLMs interpret natural language mood input + multimodal audio signals

[v] Dynamic Narrative Curation

Generate personalised playlist stories, not just song lists

[v] Conversational Discovery

User says: I need focus music for a deadline — agent understands and acts

[v] Generative Content Layer

AI-written liner notes, artist bios, mood transitions narrated intelligently

Core Insight:

Music listening is an emotional act, not just a preference act. GenAI can bridge the gap between how a user feels right now and what they actually want to hear.

[Mood] MoodSync AI

For: Mood-Driven Users (25%)

User speaks or types their current emotion. LLM interprets sentiment + context, generates a dynamic playlist with a narrative arc — from where they are emotionally to where they want to go.

[+] +22% session length

[Context] Context Conductor

For: Passive Listeners (40%)

Multimodal AI agent reads time, location (opt-in), calendar events, and ambient sound to automatically queue contextually perfect music without user input.

[+] +18% retention

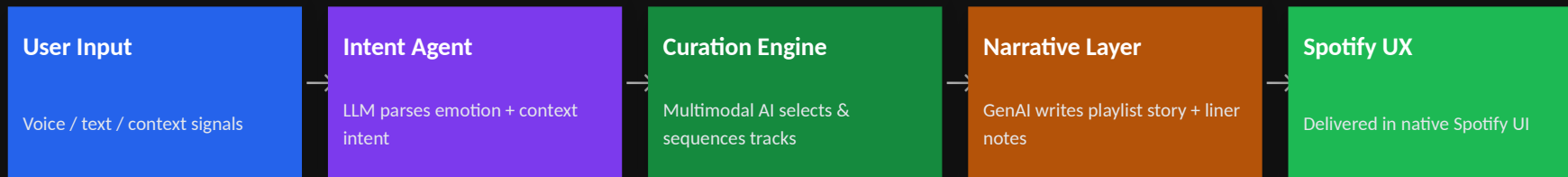
[Search] Discovery Dialogue

For: Active Explorers (35%)

Conversational AI interface: 'Show me artists like Radiohead but more hopeful.' Agent searches, reasons, and builds a curated session with AI-written discovery notes.

[+] +31% new artist streams

AI Interaction Architecture



Key Workflows & Features

[Mic] Voice Mood Check-In

"I'm anxious before a presentation" → Agent detects anxiety, generates a 20-min focus + calm arc playlist with AI narration. Post-session: asks how you feel, refines future curations.

[AI] Autonomous Session Agent

Runs in background: checks time (7am = energising commute mix), calendar ("big meeting at 10am" = confidence music), then queues without asking. Full opt-in.

[Chat] Conversational Discovery

Chat interface within Spotify: "Find something like lo-fi but with vocals and a 90s feel." LLM reasons across the catalogue, returns curated session with explainability notes.

[Write] AI Liner Notes

Each AI playlist includes a generated narrative: why these songs, what the arc is, artist context. Increases engagement and perceived value of the AI curation.

Success Metrics

+22%

Avg. session length

MoodSync AI cohort vs control

+18%

30-day retention

Context Conductor users vs baseline

+31%

New artist stream rate

Discovery Dialogue activation

+15%

Free → Premium conversion

AI features gated in Premium

-27%

Support tickets

AI resolves 'I can't find music for X' self-serve

Pitfalls & Workarounds

[!] Hallucinations

Constrain LLM to Spotify catalogue API; no free-form generation of song names. Always ground to real tracks.

[!] Privacy (context data)

Full opt-in for location/calendar. On-device inference for mood signals. GDPR-compliant data minimisation.

[!] Latency

Pre-compute context profiles async; intent agent runs on user action with <800ms SLA target using streaming responses.

[!] User Trust

Always show AI reasoning ('Here's why I picked these'). Add one-tap feedback loop. Never hide that it's AI-generated.

[!] Adoption

Progressive rollout: start with AI Liner Notes (low risk, high delight), then unlock agents for opted-in users after trust is built.